Student Performance Descriptive Analysis

Introduction

This report provides an in-depth analysis of student performance, based on data from Sling Academy, to uncover insights for better student development and strategic decision-making, using Exploratory Data Analysis and Data visualization in Excel.

Objective

The objective of this project is;

- To identify the key features that influence student performance
- To develop an interactive dashboard to monitor the real - time student performance

Dataset Description

The selected dataset for this project was “Student-Scores”, sourced from Sling Academy. This dataset includes 2000 data and 17 attributes, including;

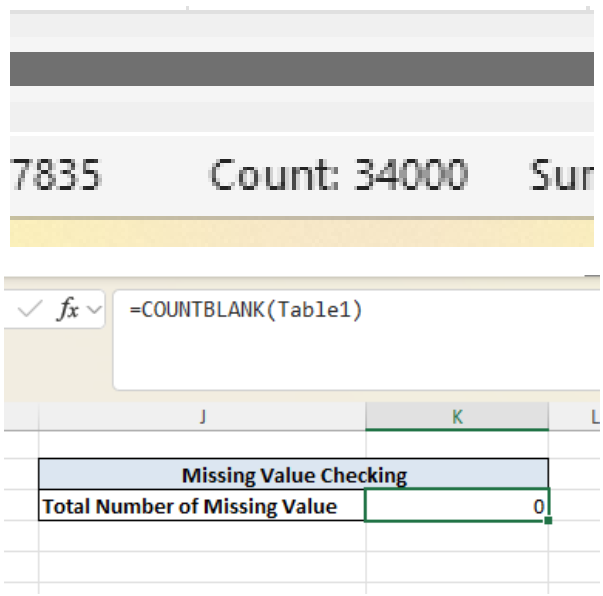
Attribute Name	Attribute Description	Data Type
ID	Unique Student Identifier	Integer
First_name	Student’s first name	String
Last_name	Student’s last name	String
Email	Student’s academy email	String
Gender	The gender of a student.	String
Math_score	Mathematics score	Float
History_score	History score (0 – 100)	Float
Physic_score	Physic score (0 – 100)	Float
Chemistry_score	Chemistry score (0 – 100)	Float
Biology_score	Biology score (0 – 100)	Float
English_score	English score (0 – 100)	Float
Geography_score	Geography score (0 – 100)	Float
Part_time_job	Do students have a part-time job or not	Boolean
Absence_days	Total number of days the students do not attend the class	Integer
Extracurricular_activities	Students join any extracurricular activities	Boolean
Weekly_self_study_hours	Total number of hours the student studies (Per Week)	Integer
Career_aspiration	Student’s career aspirations or goals for the future	String

Source:https://www.slingacademy.com/article/student-scores-sample-data-csv-json-xlsx-xml/#student-scores.xlsx

Data Preparation

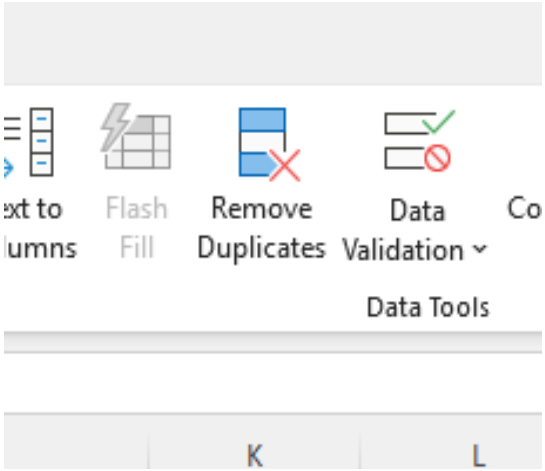
Before analysing data, data preparation is required to ensure it is valid and easier to analyse. Data preparation also allows the analyser to identify additional hidden outcomes, thereby improving data accuracy. Data preparation for this project includes checking and removing duplicate and missing values, and feature engineering.

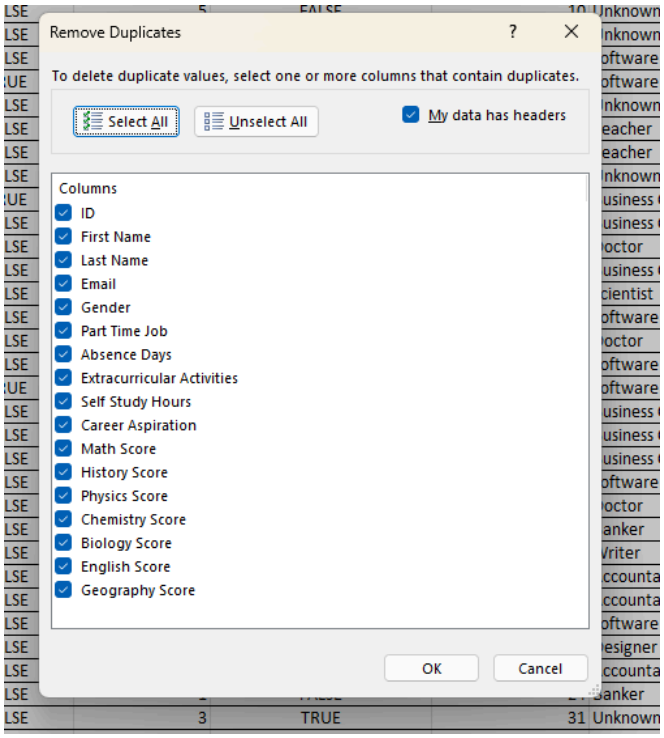
a. Identify and Remove Missing Value

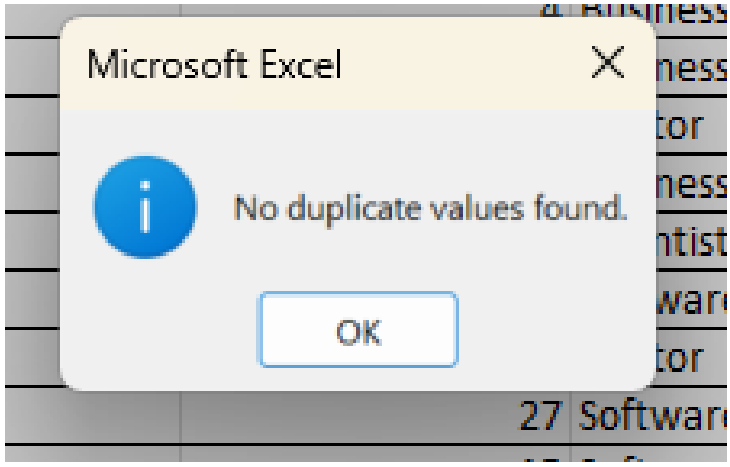


The image above shows that the total number of values in this dataset was 34000 (2000 data points × 17 columns). It can prove that there are no missing values in this dataset. Additionally, the formula “COUNTBLANK” can count the total number of missing values. The formula also proves that the number of missing values in this dataset was 0.

b. Identify and Remove Duplicate Value







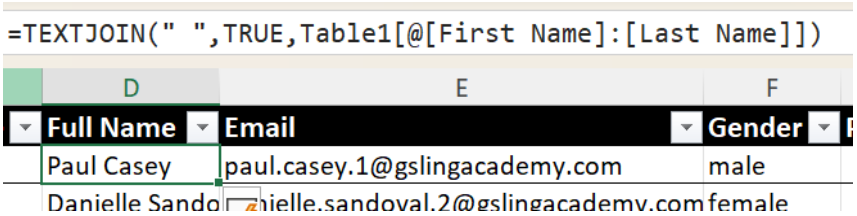
The Remove Duplicates function was applied to remove duplicate records, and the result above shows that this dataset contains no duplicates.

c. **Feature Engineering**

In this phase, new features will be created to provide more meaningful and logical insight. The new features to be created in this dataset are;

1. **Full Name**

   Formula: (=TEXTJOIN(" ",TRUE,Table1[@[First Name]:[Last Name]])



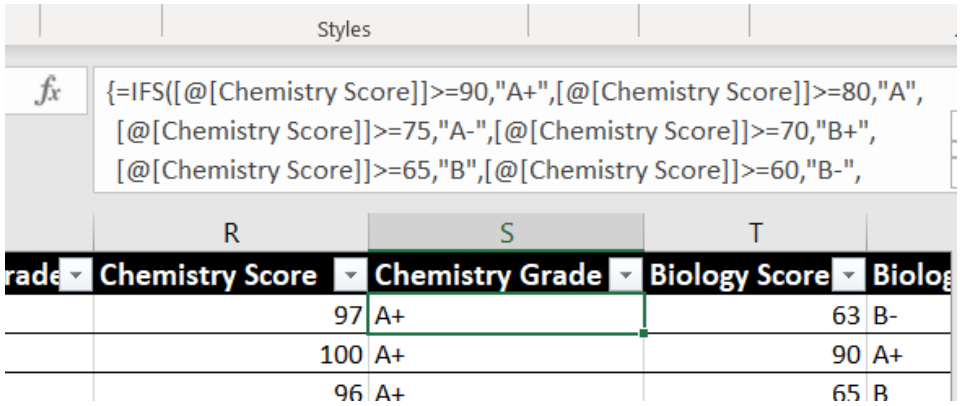
=TEXTJOIN(" ",TRUE,Table1[@[First Name]:[Last Name]])

D	E	F
Full Name	Email	Gender
Paul Casey	paul.casey.1@gslingacademy.com	male
Danielle Sando	nielle.sandoval.2@gslingacademy.com	female

2. **Grade for each subject score (Math Grade, History Grade, Physics Grade, Chemistry Grade, Biology Grade, English Grade, and Geography Grade)**

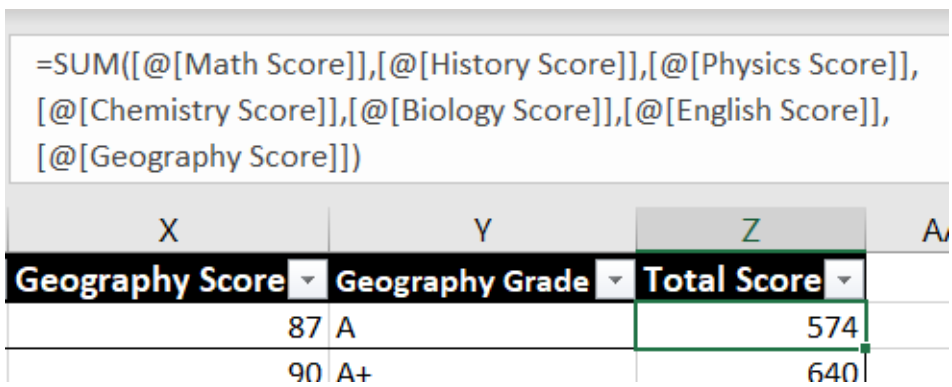
   Formula: (=IFS([@[Math Score]]>=90,"A+",[@[Math Score]]>=80,"A", [@[Math Score]]>=75,"A-",[@[Math Score]]>=70,"B+",[@[Math Score]]>=65,"B",[@[Math

Score]]>=60,"B-",[@[Math Score]]>=50,"C",[@[Math Score]]>=40,"D",[@[Math Score]]<40,"F"))



3. **<u>Total Score</u>**

Formula: SUM([@[Math Score]],[@[History Score]],[@[Physics Score]],[@[Chemistry Score]],[@[Biology Score]],[@[English Score]],[@[Geography Score]])



4. **<u>Average Score</u>**

Formula: =AVERAGE([@[Math Score]],[@[History Score]],[@[Physics Score]],[@[Chemistry Score]],[@[Biology Score]],[@[English Score]],[@[Geography Score]])

=AVERAGE([@[Math Score]],[@[History Score]],[@[Physics Score]],[@[Chemistry Score]],[@[Biology Score]],[@[English Score]],[@[Geography Score]])

	Z	AA	AB	AC	AD
rade	Total Score	Average Score			
	574	82.00			

5. **<u>Overall Grade</u>**

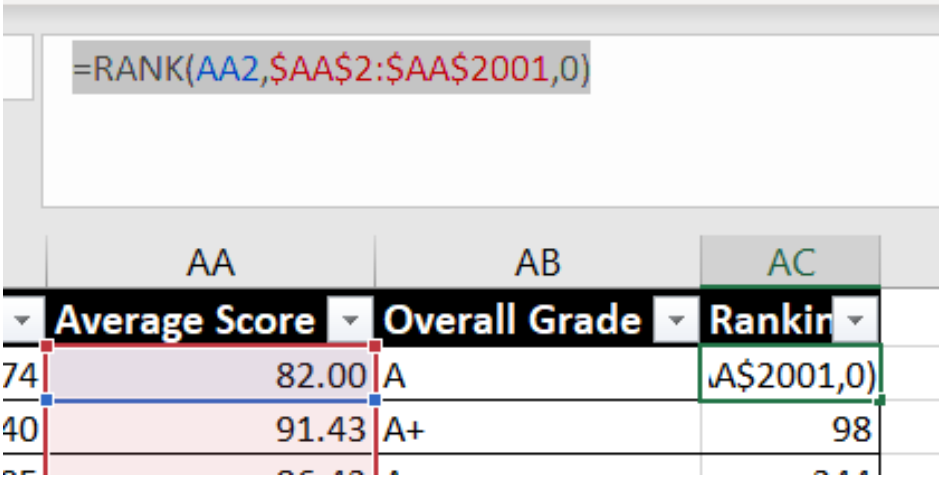
Formula: =IFS([@[Average Score]]>=90,"A+",[@[Average Score]]>=80,"A", [@[Average Score]]>=75,"A-",[@[Average Score]]>=70,"B+",[@[Average Score]]>=65,"B",[@[Average Score]]>=60,"B-",[@[Average Score]]>=50,"C",[@[Average Score]]>=40,"D",[@[Average Score]]<40,"F")

=IFS([@[Average Score]]>=90,"A+",[@[Average Score]]>=80,"A",[@[Average Score]]>=75,"A-",[@[Average Score]]>=70,"B+",[@[Average Score]]>=65,"B",[@[Average Score]]>=60,"B-",

	AA	AB	AC	AD	A
	Average Score	Overall Grade			
74	82.00	A			
40	91.43	A+			
05	86.43	A			

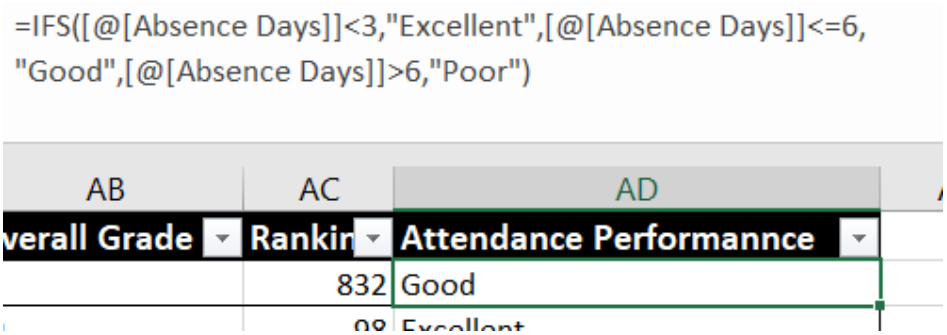
6. Ranking in the Academy

   Formula: RANK(AA2,$AA$2:$AA$2001,0)



7. Attendance Performance

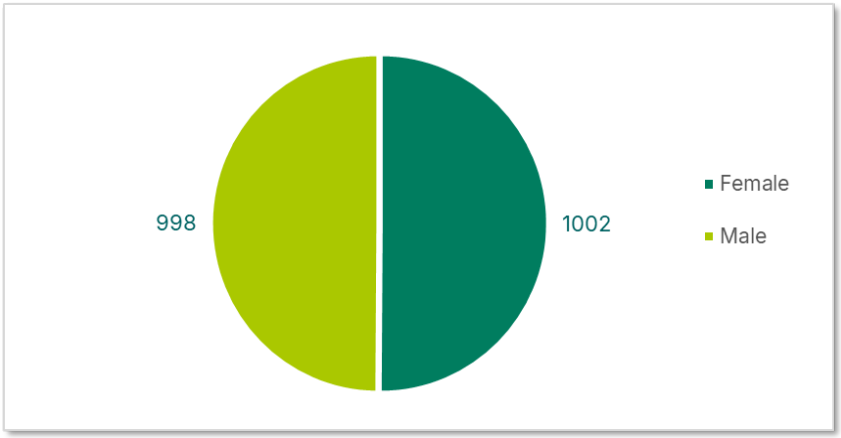
   Formula: IFS([@[Absence Days]]<3,"Excellent",[@[Absence Days]]<=6,"Good",[@[Absence Days]]>6,"Poor")



Data Finding and Analysis

After the data preparation phase, the data is considered clean and ready for analysis. In this phase, a Pivot table and a chart will be applied to generate insights from the data.

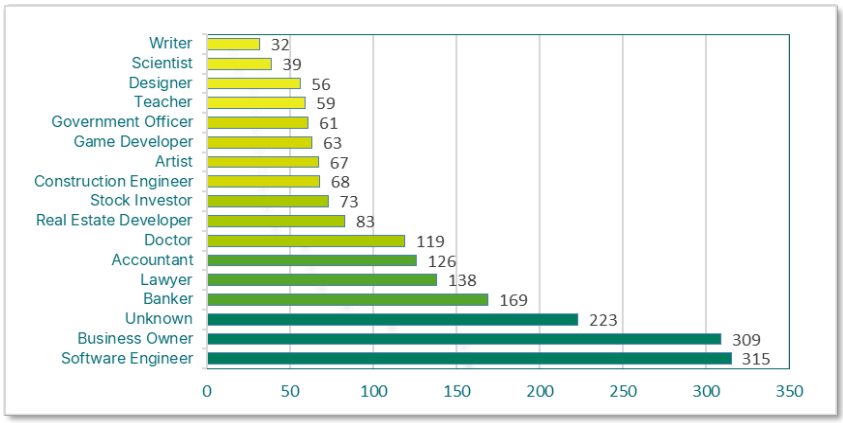
1. Gender Distribution



Row Labels	Count of Gender
Female	1002
Male	998
Grand Total	**2000**

Based on the pie chart and pivot table above, 1002 students at Sling Academy were female,, while 998 were male.

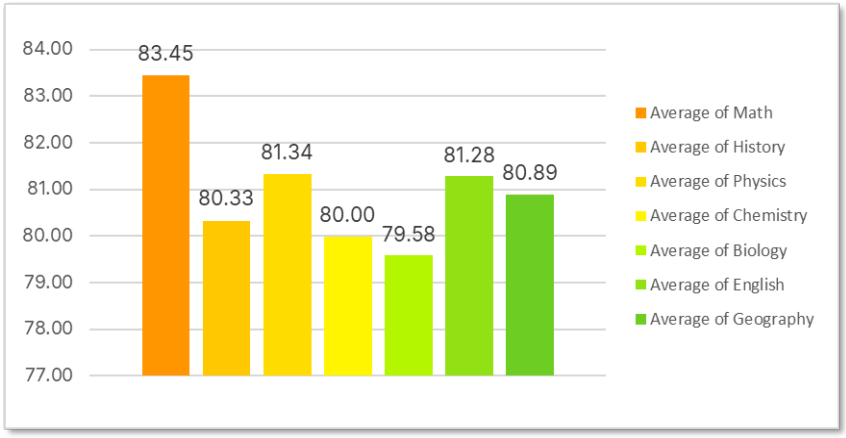
2. Career Aspiration



Software Engineer	315
Business Owner	309
Unknown	223
Banker	169
Lawyer	138
Accountant	126
Doctor	119
Real Estate Developer	83
Stock Investor	73
Construction Engineer	68
Artist	67
Game Developer	63
Government Officer	61
Teacher	59
Designer	56
Scientist	39
Writer	32
Grand Total	**2000**

According to the pivot table and horizontal bar chart, 315 students dream of becoming software engineers, followed by business owners (309), and 223 students have not yet defined their career aspirations. Sling Academy suggested organising a counselling program to provide career guidance and suggestions based on the students' performance.

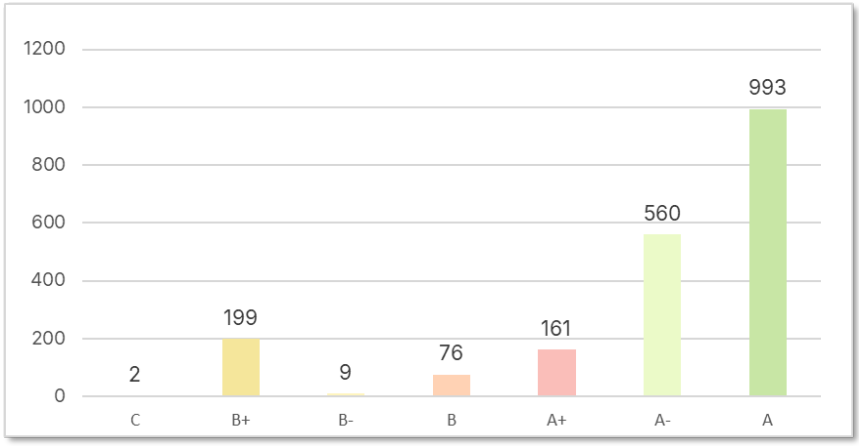
3. Subject Performance



Average of Math	Average of History	Average of Physics	Average of Chemistry	Average of Biology	Average of English	Average of Geography
83.45	80.33	81.34	80.00	79.58	81.28	80.89

In terms of subject performance, Mathematics recorded the highest average score with 83.45. Then, it's followed by Physics (81.34) and English (81.28). Biology had the lowest average score, at 79.58. It's reflected that Biology has potential challenges in content understanding and engagement.

4. Overall Grade Distribution



A	993
A-	560
A+	161
B	76
B-	9
B+	199
C	2
Grand Total	**2000**

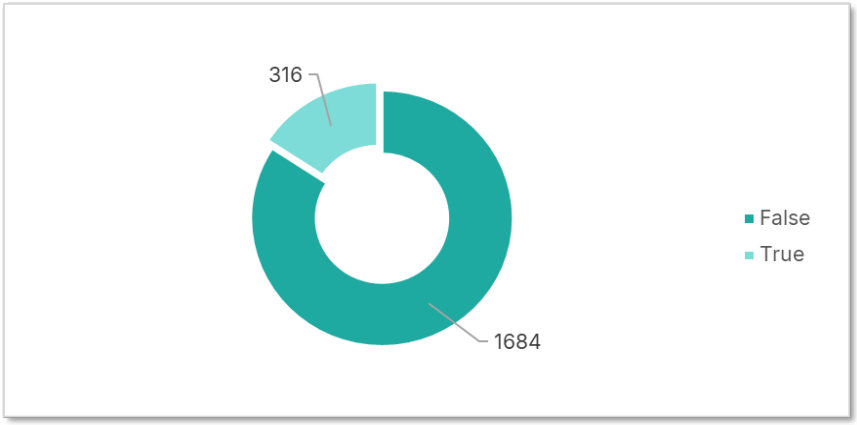
The majority of the students in Sling Academy score A (993 students) and A- (560 students) for the overall result. The lower grades, like B- and C, are minimal, with 9 and 2 students, respectively. It shows that the student's performance is more satisfactory. However, there is still room for improvement.

5. Top 10 Students Performance

Name	Average Score	Ranking
Lisa Mitchell	96.14	1
Todd Howard	95.86	2
Paula Hernandez	95.86	2
Mary Arnold	95.71	4
Paul Shaffer	95.71	4
Daniel Lewis	95.29	6
Nicholas Clayton	95.29	6
Lance Nelson	95.14	8
Scott Weaver	95.00	9
Kelly Morris	95.00	9

The academic competition at this institution is very strong, as seen by the top-performing students' scores of 95.00 to 96.14. The top students' continuously exceptional performance and the close gap in achievement at the highest level are demonstrated by the small score differences among them.

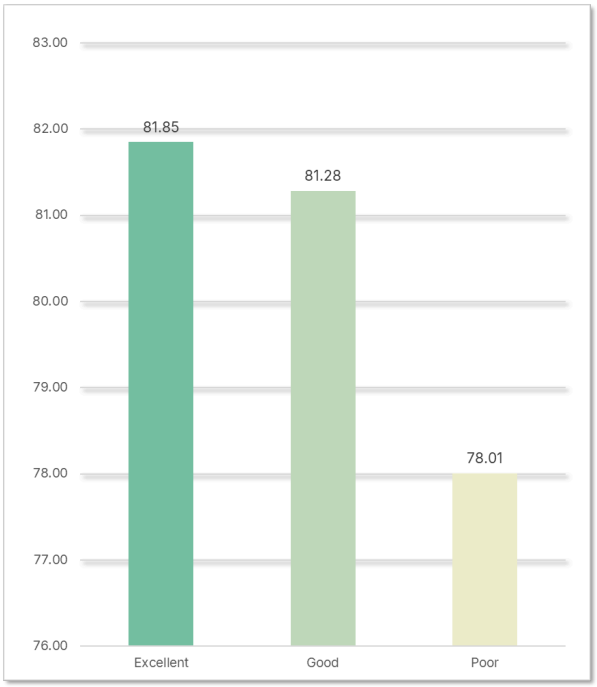
6. Part-Time Job Distribution



Row Labels	Count of Part Time Job
False	1684
True	316
Grand Total	**2000**

There are 1684 students, or 84.20% of the students do not take any part-time job, whereas 316 students or 15.8% of 2000 students, have part-time jobs. This result shows that most students indicate a strong focus on academic activities.

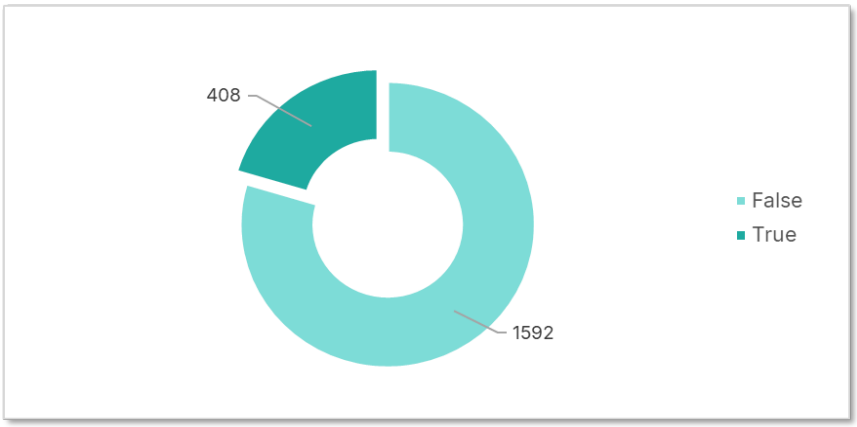
7. Attendance Performance



Row Labels	Average of Average
Excellent	81.85
Good	81.28
Poor	78.01
Grand Total	**80.98**

The highest average score of 81.85 was reached by students with Excellent attendance, closely followed by those with Good attendance (81.28). Students with poor attendance, on the other hand, had a much lower average score of 78.01.

8. Extracurricular Activities Distribution

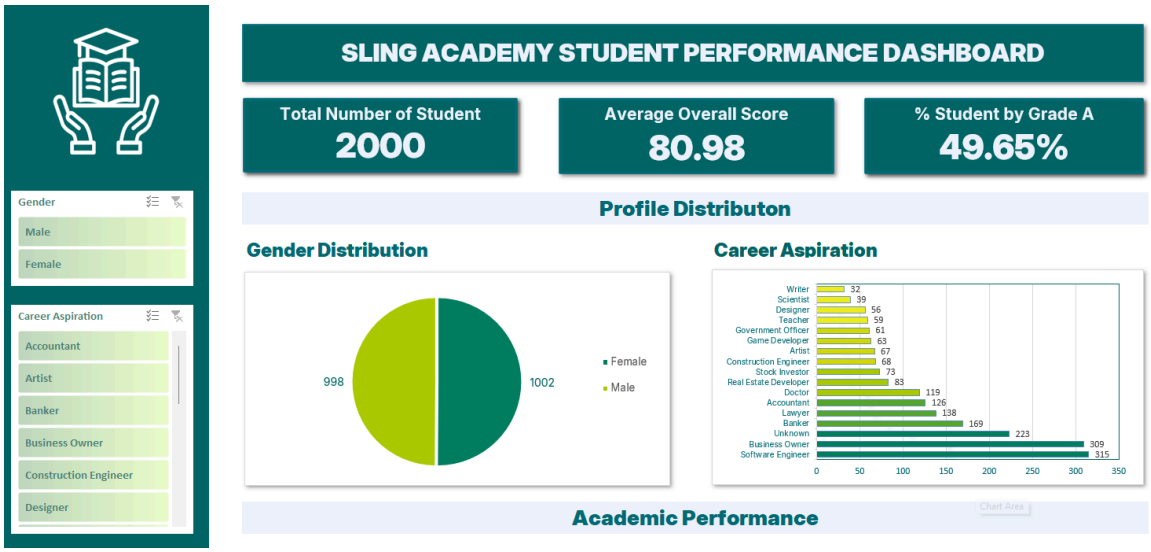


Row Labels	Count of Extracurricular Activities
False	1592
True	408
Grand Total	**2000**

There are 1592 students, or 79.60% of the students do not take any extracurricular activities , whereas 408 students or 20.40% of 2000 students, have extracurricular activities. This result shows that most students indicate a strong focus on academic activities.

Data Visualization

Finally, the results will be visualised in the dashboard in Excel, where users can view the pattern and other details. This dashboard is divided into three parts: Profile Distributions, Academic Performance, and Behaviour Analysis. 3 cards offer a quick overview of the overall interaction with student performance. Users can explore the data based on 5 criteria: gender, career aspirations, part-time job, extracurricular activities, and attendance performance. The first part (Profile Distribution) displayed the overall distribution of students' genders at Sling Academy and their career aspirations. Then, the second part, Academic Performance, illustrates the Top 10 students and the overall average grade and subject distribution to identify which students or subjects are strong or weak. The third part of this dashboard is Behaviour Analysis, which identifies the factors that affect students' performance.



Construction Engineer
Designer

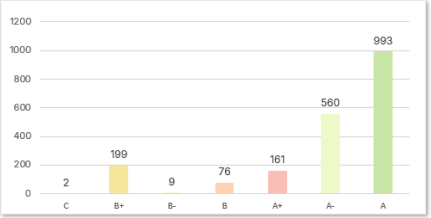
Part Time Job
True
False

Extracurricular Activities
TRUE
FALSE

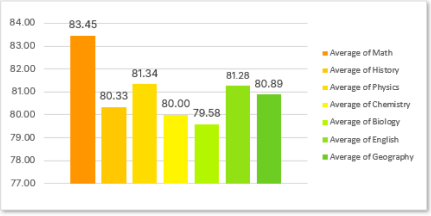
Attendance Performan...
Excellent
Good
Poor

Academic Performance

Overall Grade Distribution



Subject Performance



Top 10 Student

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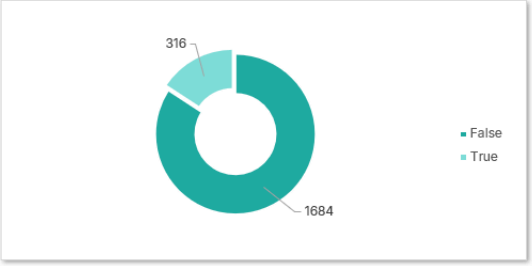
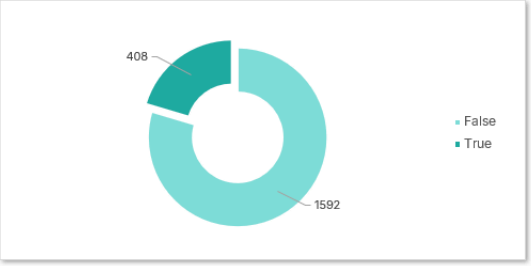
Chart Area

Behavior Analysis

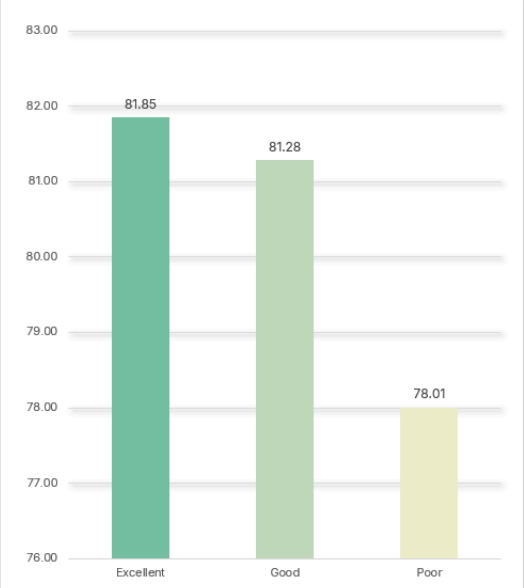
Part-time Job Distribution



Extracurricular Activities Distribution



Attendance Performance



Recommendations

1. Tutorial programs - The institution is encouraged to conduct targeted tutorial or review classes for low-performing students and those with poor attendance. These programs help students catch up on missed lessons, strengthen their understanding of key concepts, and improve overall academic performance.
2. Counselling & Support Services - Teachers or administrators need to engage with students who consistently miss class to identify the reasons behind the absences and provide mental support, whether academic or personal. It can help address these issues and improve both attendance and performance.
3. Increase engagement in extracurricular activities: The university should create and promote a broader range of activities to encourage greater student participation. This will provide students with the opportunity to explore varied interests, develop soft skills such as teamwork and leadership, and cultivate beneficial habits that contribute to both personal growth and academic performance.